TEXORA · LECTURE SHEET

Knitted Fabric Production Calculations

Sheet 8 produced a rate for one machine. Production calculation turns rates into commitments: an order in kilograms becomes machine-shifts, machine-shifts become machines and days, and the yarn store is told exactly what to hold — with waste and stoppages priced in before they happen, not after.

FM-3207 · ADVANCED WEAVING AND KNITTING CALCULATIONS · TOPIC 10 · SPECIALIZATION — FABRIC MANUFACTURING

1. The reference rate

From Sheet 8: the 30-inch, 24-gauge, 90-feeder machine at 20 rpm, stitch length 2.9 mm, Ne 30, produces **94.8 kg per 8-hour shift at 85% efficiency** of 132 GSM jersey, 1.89 m open width. Every calculation in this sheet stands on that one verified rate — production planning is only ever as good as the rate it multiplies.

2. Order to machine-shifts to days

WORKED EXAMPLE — A 25-TONNE ORDER

Order: **25,000 kg** of the Sheet 8 fabric.

Machine-shifts required = 25,000 ÷ 94.8 = 263.7 → **264 machine-shifts**.

The shed runs 18 such machines, 3 shifts per day = 54 machine-shifts per day.

Days = 264 ÷ 54 = **4.9 days** — promise the buyer day 6, not day 5: the 0.9 is where breakdowns live.

In cloth terms the order is 25,000 ÷ 0.1322 = **189,107 m²**, i.e. 189,107 ÷ 1.89 ≈ **100,057 m** of fabric — the figure the finishing and inspection departments plan against.

3. Yarn requirement with waste

Knitting waste — cone tails, start-up fabric, faulty courses cut out, oil-stained ends — is small but never zero; 2% is a working allowance for a well-run jersey shed.

$$\text{Yarn to purchase} = \text{fabric weight} \times (1 + \text{waste allowance})$$

WORKED EXAMPLE — YARN FOR THE 25-TONNE ORDER

Yarn = 25,000 × 1.02 = **25,500 kg** of Ne 30.
In cones of 1.36 kg (Sheet 1's package) that is 25,500 ÷ 1.36 ≈ 18,750 cones — a number the store, not the knitters, must feel in advance. If the winding is in-house, Sheet 1's spindle arithmetic converts it again into winding capacity; the chain order → fabric → yarn → packages → spindles is the same chain Sheet 6 built for weaving.

4. Efficiency as a measured quantity

The 85% in the rate is not a hope; it is built from the stoppage record. A machine stopping 6 times per hour for an average of 2.5 minutes loses 15 of every 60 minutes — 25% of the hour — before any other loss is counted; a machine with the same stop count at 1 minute per stop loses only 10%. This is why production calculation distinguishes:

- **rpm efficiency** — actual courses against the courses the set rpm would give;
- **time efficiency** — running hours against attended hours;
- **the plan rate** — measured kg per shift over the last comparable order, which quietly contains both.

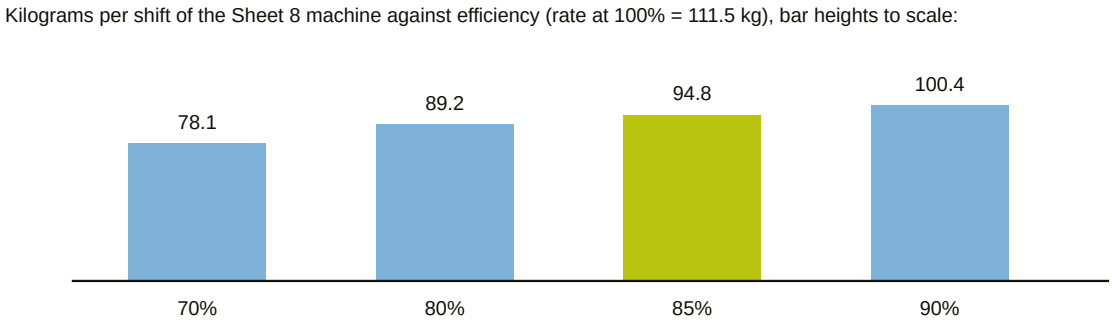


Figure 1 — Efficiency is a multiplier on money: five points of it are worth 5.6 kg on every shift this machine ever runs.

5. Garment-length and panel thinking

Where fabric is knitted to body-size tubes or garment panels, the same arithmetic is run per piece: courses per panel = panel length × CPC; yarn per panel = courses × yarn per course; panels per shift = machine output ÷ yarn per panel. The discipline is identical — only the unit changes from the kilogram to the dozen — and the waste allowance grows, because cutting waste joins knitting waste. The planner who can run Sections 2–3 can run this; the planner who cannot, cannot.

Glossary

Machine-shift — one machine worked for one shift; the currency of knitting capacity. **Plan rate** — measured production per shift from comparable past work. **Waste allowance** — % added to fabric weight to obtain yarn purchase weight. **Stoppage record** — stops per hour and minutes per stop, kept separately.

Key points

• Machine-shifts = order ÷ rate per shift (worked: 264 shifts; 18 machines × 3 shifts → 4.9 days — promise day 6).
• Yarn purchase = fabric × (1 + waste): 25,000 kg → 25,500 kg ≈ 18,750 cones.
• Orders are dual-currency: 25,000 kg = 189,107 m² = ≈100,057 m of this cloth.
• Efficiency is measured from stoppages, and it multiplies every plan built on it.

Practice

1. How many machines are needed for 9,000 kg in 5 days at 2 shifts per day, on a machine producing 95 kg per shift?
2. A machine's stoppage record improves from 6 stops/h × 2.5 min to 6 stops/h × 1.5 min. What happens to its stopped fraction, and qualitatively to its plan rate?
3. Express a 12,000 kg order of Sheet 12's fabric (179 GSM, 1.76 m wide) in metres. (Sheet 12 obtains 38,069 m — check your route against it.)

Answer key

1. Machine-shifts = 9,000 ÷ 95 = 94.7 → 95; shifts available per machine = 5 × 2 = 10; machines = 95 ÷ 10 = 9.5 → **10 machines**.
2. Stopped time falls from 15 to 9 minutes per hour — the stopped fraction drops from 25% to 15%, and the plan rate rises in proportion to the recovered running time (about +13% on running rate): same stops, shorter stops, real money.
3. Area = 12,000 ÷ 0.1791 = 67,002 m²; length = 67,002 ÷ 1.76 = **38,069 m ✓**.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib